DB-Curator

Agentic scientific database construction with auditable evidence

Dynamic literature construction + predicate-state verification

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Why scientific database construction is an agentic task

Not Q&A

Task mismatch

A database row must be accepted, rejected, or left evidence-limited with provenance.

Dynamic

Corpus construction

The paper set is built during the run from scope, candidates and critique feedback.

Auditable

Entry-level evidence

Accepted compounds need cited papers, local context and evidence spans.

less manual triage | **explicit inclusion/exclusion** | **paper-grounded rows** | **edge-case review**

Core message for ML + science: the agent is useful because it controls search, state, and stopping criteria, not because it writes fluent summaries.

DB-Curator workflow

1 Scope

request → rubric
inclusion / exclusion

2 Literature

broad search
seed expansion
follow-up

3 Candidates

NER + grounding
ranking + queue

4 Verification

Researcher reads
Critique audits

5 Database

accepted rows
spans + IDs

Predicate-state loop

supported | unresolved | exclusion | exhausted

Critique asks which predicate is unsupported, what evidence is missing, and what should be searched next.

Application: release-ready database rows

178

Known entries

68

Benchmark-missing valid

4

Low release error

Release audit over five domains



Rows out of 250: benchmark overlap, benchmark-missing true, false

Scientific punchline: the system recovers known entries and surfaces literature-buried additions with auditable evidence.

Verification: evidence trails survive paper-level audit

2,166/2,166

DB-Curator release audit

All cited primary-paper references contained supporting span or local context.

11/257

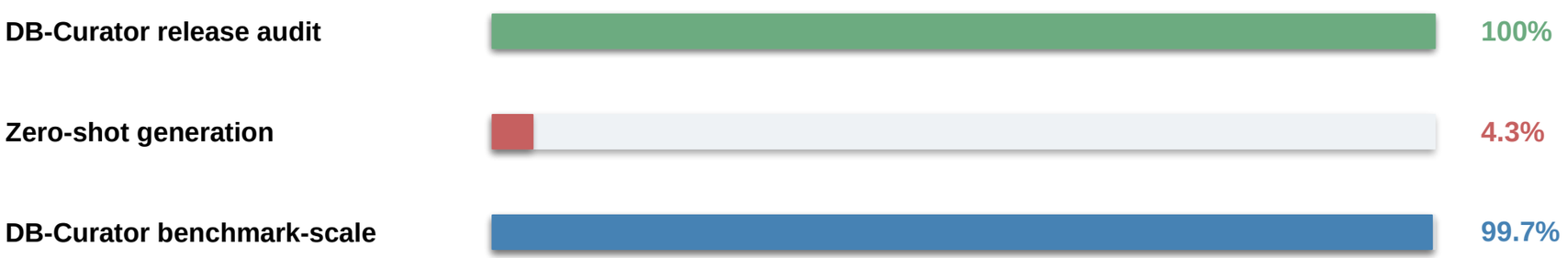
Zero-shot generation

Plausible rows, but few usable evidence trails.

7,430/7,453

Benchmark-scale audit

Almost all cited references had quote/context support.



Audit rule: resolve paper identity → search recovered text → accept exact quote or equivalent local context → exclude lookup pages.

Benchmark-scale verification across domains

1,887

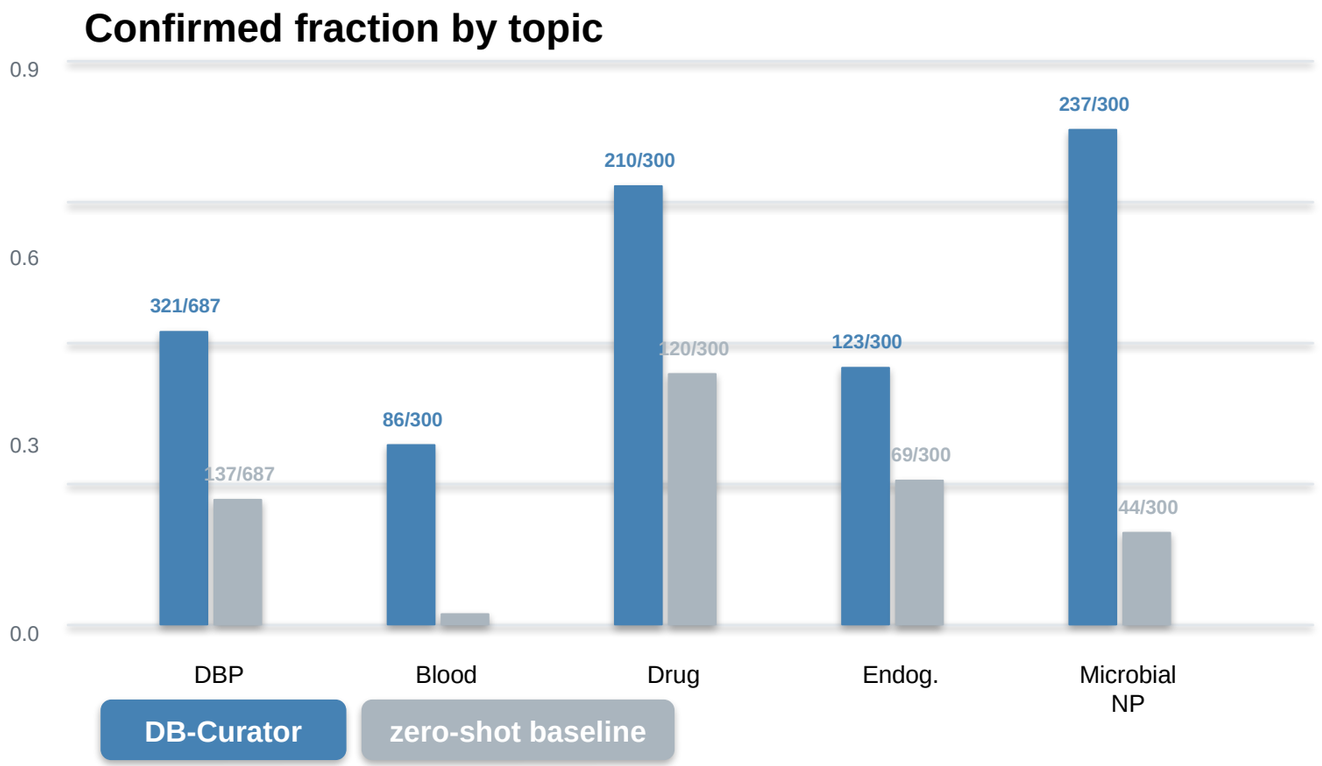
Candidate rows

977

Confirmed by system

375

Baseline



ML takeaway: this is not one classifier threshold. Different topic definitions create different predicate gates and failure modes.